Dr. Manpreet Singh

Assistant Professor, Department of Chemistry, Saroop Rani Government College for Women, Amritsar, India–143001

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Education

Guru Nanak Dev University, Amritsar, India
Ph.D. Chemistry 05/09/2019 – 21/01/2025
Degree Awarded on 21/01/2025

Guru Nanak Dev University, Amritsar, India
M.Sc. Chemistry 01/07/2016 – 01/05/2018
Degree Awarded on 08/06/2018

Guru Nanak Dev University, Amritsar, India
B.Sc. Chemistry 01/07/2012 – 01/05/2015
Degree Awarded on 08/07/2015

NET-JRF

Qualified **CSIR-UGC JRF** with AIR 42 09/08/2019

Research Experience

Guru Nanak Dev University, Ph.D. Chemistry 05/09/2019 – 21/01/2025
Research Objective: To enhance the rate of different enzymatic reactions by using aqueous solutions of diverse surface-active ionic liquids as fluid scaffolds at various concentrations.

The researcher has worked on the synthesis of various cationic and anionic Surface active ionic liquids (SAILs). The synthesized SAILs were investigated for self-assembly in aqueous medium using numerous techniques, which include but not limited to conductivity measurements, fluorescence spectroscopy, surface tension measurements, small angle neutron scattering (SANS) etc. Further, the aqueous solutions of SAILs at different concentrations have been used as medium to surge the rate of various enzymatic reactions.

Doctoral Dissertation Title: Studies on Enzymology in Aqueous Solutions of Surface-Active Ionic Liquids.

Guru Nanak Dev University, Masters Research Project 01/07/2017 – 31/12/2018
Supervisor: Prof. Tejwant Singh Kang

Research Objective: To investigate the various excess properties such as excess molar volume, partial molar volume, excess partial molar volume etc. of binary mixture of Ethyl ammonium nitrate with the derivatives of diethylene glycol.

Master Thesis title: Density and Excess Properties of a Protic Room Temperature Ionic Liquid (Ethyl ammonium nitrate) with Diethylene Glycol and Its Derivatives at T = (298.15 to 318.15 K)

Peer-Reviewed Publications

1. Manpreet Singh, Rajwinder Kaur, Sugam Kumar, Vinod Kumar Aswal, Gurbir Singh* and Tejwant Singh Kang*. Self-assembly of 2-hydroxyethyl-1H-imidazolium based Surface Active Ionic Liquid and Utilization of Their Aqueous Solution in Super-activity of Cytochrome-c, *J. Phys. Chem. B*, 2026, 129, 26, 6661–6673.

2. Harmandeep Kaur,Rajwinder Kaur, Muskan, Manpreet Singh, Ravi Dutt, Kanica Sharma, Harjinder

Singh, Kuldeep Singh, Arvind Kumar, Gurbir Singh and Tejwant Singh Kang*. Unprecedently high dissolution and depolymerization of hard-to-dissolve polyurethane in deep eutectic solvents under low-energy white light, *RSC Sustainability*, 2026 4, 952–961.

3. Nipunn Sharma, Surbhi Sharma, Pretam Kumar, Manpreet Singh, Vivek K. Gupta and Sushil K. Pandey*. Structure-property correlations of Ce(III) and Sm(III) phosphorodithioates: Experimental and DFT perspectives, *Journal of Molecular Structure*, 2026, 1377, 147098.

4. Nipunn Sharma, Pretam Kumar, Anu Radha, Manpreet Singh, Tejwant Singh and Sushil Kumar Pandey*. Comprehensive molecular profiling of Nd(III) complexes of diaryldithiophosphates: An integrated spectroscopic, magnetic, thermal, and computational study. *Polyhedron*, 2026, 283, 117846–117854.

5. Manvir Kaur, Manpreet Singh, Rajwinder Kaur, Navdeep Kaur, Pratap Kumar Pati and Tejwant Singh Kang. Surfactant-free microemulsion as a fluid scaffold for the thermal stabilization of lysozyme. *Phys. Chem. Chem. Phys.*, 2025, 27, 10580–10590.

6. Manish Sharma, Palak Khurana, Manpreet Singh, Tejwant Singh Kang and Inderpreet Kaur*. Synthesis, characterisation and application of thiol-based spectroscopic and electrochemical sensors for detection of iron(III) in environmental samples – A DFT supported comparative study. *Int. J. Environ. Anal. Chem.*, 2025, 1–17.

7. Tejwant Singh Kang*, Masa-aki Morikawa, Manpreet Singh and Nobuo Kimizuka*. Electric Field-Driven Long-Range Order and Enhanced Polarization Switching in High-Dipole Ionic Liquids. *J. Am. Chem. Soc.*, 2025, 147, 8809–8819.

8. Manpreet Singh, Gurbir Singh, Harmandeep Kaur, Muskan, Sugam Kumar, Vinod Kumar Aswal and Tejwant Singh Kang*. Self-assembly of choline-based surface-active ionic liquids and concentration-dependent enhancement in the enzymatic activity of cellulase in aqueous medium, *Phys. Chem. Chem. Phys.*, 2024, 26, 16218–16233.

9. Richu, Qammer Majid, Himani Singh, Akshita Bandral, Manpreet Singh, Tejwant Singh Kang and Ashwani Kumar*. Solution Properties of Folic Acid/Nicotinic Acid in Aqueous 1-Butyl-3-methylimidazolium-chloride Media at Discrete Concentrations and Temperatures: A Thermodynamic, Computational, and Spectroscopic Study. *J. Chem. Eng. Data*, 2024, 69, 2503–2527.

10. Harmandeep Kaur, Manpreet Singh, Navdeep Kaur, Pratap Kumar Pati, Monika Rani and Tejwant Singh Kang*. Sustainable Dissolution of Collagen and Formation of Polypeptides in Deep Eutectic Solvents for Use as Antibacterial Agents. *RSC Sustainability*, 2024, 2, 2312.

11. Manjot Kaur, Gurbir Singh, Riya Shivgotra, Manpreet Singh, Shubham Thakur and Subheet Jain. Prolonged Skin Retention of Luliconazole from SLNs Based Topical Gel Formulation Contributing to Ameliorated Antifungal Activity. *AAPS PharmSciTech*, 2024, 25(7), 229.

12. Harjinder Singh, Kanica Sharma, Karthikeyan Sekar, Manpreet Singh, Tejwant Singh Kang. Biosurfactant mediated sustainable exfoliation of MoS2 and in-situ preparation of Ag@MoS2 nanocomposites for photocatalysis under sunlight in aqueous medium. *Surfaces and Interfaces*, 2024, 48, 104272.

13. Manpreet Singh, Sugam Kumar, Vinod Kumar Aswal and Tejwant Singh Kang*. Mixed Aggregates of Surface-Active Ionic Liquids and 14-2-14 Gemini Surfactants in an Aqueous Medium as Fluid Scaffolds for Enzymology of Cytochrome-c, *Langmuir*, 2023, 39, 11582–11595.

14. Jyoti Sharma, Manpreet Singh, Tarlok Singh Banipal, Tejwant Singh Kang and Parampaul Kaur Banipal*. Evaluating the binding ability of promethazine hydrochloride drug with sodium do-

decylsulphate/sodium lauroylsarcosinate/sodium cholate in the company of uridine biomolecules: experimental and theoretical approach. *Journal of Dispersion Science and Technology*, 2023, 1–18.

15. Manjot Kaur, Riya Shivgotra, Shubham Thakur, Rasdeep Kour, Manpreet Singh, Navid Reza Shahtaghi, Satwinderjeet Kaur and Subheet Kumar Jain*. A novel nanoemulgel formulation of Luliconazole with augmented antifungal efficacy: In-vitro, in-silico, ex-vivo, and in-vivo studies. *Journal of Drug Delivery Science and Technology*, 2023, 89, 105102.

16. Himani Singh, Qammer Majid, Richu, Manpreet Singh, Tejwant Singh Kang, and Ashwani Kumar*. Explorations on Solute-Solvent Interactions of Tripotassium Citrate and Sodium Benzoate in Aqueous 1-Ethyl-3-methylimidazolium Ethyl Sulfate Solutions: Physicochemical, Spectroscopic, and Computational Approaches. *J. Chem. Eng. Data*, 2023, 68, 2563–2584.

17. Omish Sethi, Manpreet Singh, Ashwani Kumar Sood* and Tejwant Singh Kang*. Water Induced Alterations in Self-Assembly of a BioSurfactant in Deep Eutectic Solvent for Enhanced Enzyme Activity. *ChemPhysChem*, 2023, e202300293 (1 of 13).

18. Harmandeep Kaur, Manpreet Singh, Kuldeep Singh, Arvind Kumar, and Tejwant Singh Kang*, Sustainable Preparation of Oxidized Graphitic Material from Wheat Straw Using a Deep Eutectic Solvent for Superactivity of Cellulase. *Green Chem.*, 2023, 25, 5172–5181.

19. Manvir Kaur, Manpreet Singh, Gurbir Singh, Amritpal Singh, Gurleen Kaur, Surinder Kumar Mehta and Tejwant Singh Kang*, Water-pluronic-ionic liquid based microemulsions: Preparation, characterization and application as micro-reactor for enhanced catalytic activity of Cytochrome-c. *Colloids and Surfaces B: Biointerfaces*, 2023, 222, 113034.

20. Harmandeep Kaur, Manpreet Singh, Harjinder Singh, Manvir Kaur, Gurbir Singh, Karthikeyan Sekar and Tejwant Singh Kang*, Zinc Chloride Promoted Inimitable Dissolution and Degradation of Polyethylene in a Deep Eutectic Solvent under White Light. *Green Chem.*, 2022, 24, 2953–2961.

21. Manpreet Singh, Harjinder Singh, Omish Sethi, Ashwani Kumar Sood and Tejwant Singh Kang*. Investigating the thermodynamic properties and interactional behaviour of a protic room temperature ionic liquid in binary mixtures with ethylene glycol derivatives at different temperatures. *J. Mol. Liq.*, 2022, 367, 120410.

22. Pallavi Sohal, Rupinder Kaur, Manpreet Singh, Tarlok Singh Banipal, Parampaul Kaur Banipal* and Tejwant Singh Kang*. Binding studies of dopamine HCl drug with the mixed {sodium bis(2-ethylhexyl) sulfosuccinate + sodium dodecylsulfate} micelles: Physicochemical, spectroscopic, calorimetric and computational approach. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 2022, 640, 128386.

23. Manpreet Singh, Amrit Kaur, Harjinder Singh, Raman Kamboj, Sukhprit Singh* and Tejwant Singh Kang*. Modulation of micellization behavior of imidazolium based surface active ionic liquids by aromatic anions in aqueous medium. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 2021, 130, 127588.

24. Maninder Kaur, Muskan Dhaliwal, Harpreet Kaur, Manpreet Singh, Sneh Punia Bangar, Manoj Kumar and Ravi Pandiselvam. Preparation of antioxidant-rich tricolor pasta using microwave processed orange pomace and cucumber peel powder: A study on nutraceutical, textural, color, and sensory attributes. *J. Texture Stud.*, 2021, 1–10.

25. Gagandeep Singh, Manvir Kaur, Manpreet Singh, Harmandeep Kaur and Tejwant Singh Kang*, Spontaneous Fibrillation of Bovine Serum Albumin at Physiological Temperatures Promoted by Hydrolysis-Prone Ionic Liquids. *Langmuir*, 2021, 37(34), 10319–10329.

26. Omish Sethi, Manpreet Singh, Tejwant Singh Kang* and Ashwani Kumar Sood*, Volumetric and compressibility studies on aqueous mixtures of deep eutectic solvents based on choline chloride and carboxylic acids at different temperatures: Experimental, theoretical and computational approach. *J. Mol. Liq.*, 2021, 340, 117212.

27. Ashutosh Sharma, Manpreet Singh, Komal Arora, Prit Pal Singh, Rahul Badru, Tejwant Singh Kang* and Sandeep Kaushal*. Preparation of cellulose acetate-Sn(IV) iodophosphate nanocomposite for efficient and selective removal of Hg^{2+} and Mn^{2+} ions from aqueous solution. *Environmental Nanotechnology, Monitoring & Management*, 2021, 16, 100478.

28. Manvir Kaur, Harmandeep Kaur, Manpreet Singh, Gagandeep Singh and Tejwant Singh Kang*, Biamphiphilic Ionic Liquid Based Aqueous Microemulsions as Efficient Catalytic Medium for Cytochrome c. *Phys. Chem. Chem. Phys.*, 2021, 23, 320–328.

29. Gagandeep Singh, Manvir Kaur, Gurbir Singh, Komal Arora, Manpreet Singh, Bilal A. Sheikh and Tejwant Singh Kang*. Concentrated aqueous dispersions of low-defect few-layer thick graphene using surface active ionic liquid for enhanced enzyme activity. *Mater. Adv.*, 2020, 1, 1364.

30. Gurbir Singh, Komal, Manpreet Singh, Ormanpreet Singh and Tejwant Singh Kang*. Hydrophobically Driven Morphologically Diverse Self-Assembled Architectures of Deoxycholate and Imidazolium-Based Biamphiphilic Ionic Liquids in Aqueous Medium. *J. Phys. Chem. B*, 2018, 122, 1227–12239.

Awards and Scholarships

Young Scientist Award by Indian Council of Chemists — 28/12/2022
Senior Research Fellowship by CSIR — 05/09/2021 – 04/09/2024
Junior Research Fellowship by CSIR — 05/09/2019 – 04/09/2021

Leadership and Mentoring

Priya Sharma — 01/06/2020 – 31/05/2021
Master Thesis student, Guru Nanak Dev University, Amritsar, India
Thesis title: Hydration and Ion Association Behaviour of an Ionic Liquid 1-Butyl-3-Methyl Imidazolium Chloride: FTIR and ^{1}H NMR Study

Deepali Sharma — 01/06/2021 – 31/05/2022
Master Thesis student, Guru Nanak Dev University, Amritsar, India
Thesis title: Density, Excess Properties, Speed of Sound and Isentropic Compressibility studies of Deep Eutectic Solvent (Acetic acid + Choline Chloride) with Methanol and Ethanol at Temperature of T = 298.15 to 318.15 K

Kashish Mahajan — 01/06/2021 – 31/05/2022
Master Thesis student, Guru Nanak Dev University, Amritsar, India
Thesis title: Density, Excess Properties, Speed of Sound and Isentropic Compressibility Studies of Deep Eutectic Solvent Lactic acid + Zinc Chloride (4:1 & 3:1) with Water at Temperature of T = 298.15 to 318.15 K

Diksha — 01/06/2021 – 31/05/2022
Master Thesis student, Guru Nanak Dev University, Amritsar, India
Thesis title: Speed of Sound and Isentropic Compressibility studies on the mixture of deep eutectic solvents (Choline Chloride and Lactic Acid in 1:2) with methanol at T = (298.15 K to 318.15 K)

Shefali Chopra — 01/06/2021 – 31/05/2022
Master Thesis student, Guru Nanak Dev University, Amritsar, India
Thesis title: Density and Excess Properties of Deep Eutectic Solvent (Lactic acid + Choline Chloride) with Methanol at temperature of T = 298.15 to 318.15 K

Tanu Choudhary 01/06/2022 – 31/05/2023
Master Thesis student, Guru Nanak Dev University, Amritsar, India
Thesis title: Density and Excess Properties of Deep Eutectic Solvent (Acetic acid + Choline Chloride) with Methanol and ethanol at temperature of (298.15 to 318.15) K

Rekha Devi 01/06/2022 – 31/05/2023
Master Thesis student, Guru Nanak Dev University, Amritsar, India
Thesis title: Acacia Catechu Capped Zinc Oxide Nanoparticles Synthesis, Characterization and Application

Karanvir Singh 01/06/2022 – 31/05/2023
Master Thesis student, Guru Nanak Dev University, Amritsar, India
Thesis title: Hearding of Oil Slicks in Sea Water using Surface Active Ionic Liquids

Area of Expertise

- Ionic Liquids
- Deep Eutactic Solvents
- Microemulsions
- Enzymology
- Density Functional Theory (DFT)
- Molecular Dynamics (MD)
- Computational Chemistry

Teaching Experience

Assistant Professor, S. R. Government College for Women, Amritsar 28/09/2024 – Till date
For two semesters, The Researcher taught Physical chemistry to Second-year undergraduates, covering both Organic and Physical chemistry. Students in the 12th grade were taught general chemistry.

Guru Nanak Dev University, Amritsar, India 01/07/2022 – 31/12/2022
For one semester, The Researcher taught Quantum Mechanics to the third-year undergraduates.